Sahaj Bhadja

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Summary

Backend software engineer with 2+ years building scalable web services, REST APIs, and high-volume data workflows. Strong in Python, JavaScript/TypeScript, SQL, AWS, S3, relational databases, Docker, CI/CD, testing, and production troubleshooting.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, OOP

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected Azure SQL and Blob-backed services for 500+ users, optimizing indexed queries, schemas, and API access paths to reduce dashboard latency from 1.2 seconds to 350 milliseconds across production workflows.
- Built fault-tolerant background services for 1,000+ monthly jobs using explicit states, retries, validation, and failure isolation, improving throughput 3x while strengthening observability, debugging, and recurring production support.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Flask REST APIs and React workflows for secure file ingestion and data review, supporting 10,000+ daily requests while maintaining clean interfaces, documented services, and reliable full-stack behavior.
- Developed multithreaded processing and MySQL on AWS RDS for 1,000+ uploads per batch, reducing turnaround by 60% while improving testability, operational stability, and troubleshooting across concurrent production workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python data pipelines processing 100,000+ autonomous-driving trajectory logs, implementing validation, feature preparation, and repeatable experiment workflows that improved long-horizon planning performance by 18% across simulation benchmarks.
- Developed spatial-temporal processing with map geometry, trajectory context, and multi-head attention, improving closed-loop stability by 15% while documenting experimental behavior and debugging complex production data-dependent model failures.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform microservices and 20+ automated unit and integration suites using Jenkins, PyTest, and Mockito, improving code-review confidence, release reliability, and debugging speed across Windows, Linux, and macOS.
- Optimized REST API and ETL services with OOP patterns, AWS Glue, S3, and SQL processing, improving response times by 30% and pipeline efficiency by 68% for production data operations.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN platform with REST APIs, WebSockets, and modular Node.js services, supporting 2M+ synchronized records while improving database access, real-time updates, and full-stack coordination across distributed users.
- Automated Docker-based CI/CD testing and deployments with Jenkins, cutting release time by 50% while improving build consistency, regression coverage, and troubleshooting for frequent service changes and production rollouts.

Skills

Languages: Python, JavaScript, TypeScript, Java, SQL, C++, Bash; **Backend & APIs**: Flask, FastAPI, Node.js, Express.js, REST APIs, OpenAPI, microservices, background workers, OOP; **Frontend**: React.js, Next.js, HTML5, CSS3
Data & Cloud: MySQL, PostgreSQL, Azure SQL, SQL Server, MongoDB, AWS RDS, Redshift, Athena, S3, AWS Glue, Azure; **Engineering**: Git, Docker, Jenkins, GitHub Actions, CI/CD, PyTest, Mockito, unit testing, integration testing, code reviews, observability, debugging, documentation, Agile/Scrum

Projects

Chat2SQL Code | *Java, TypeScript, SQL, MySQL, PostgreSQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, accelerating query authoring, schema exploration, and data management through a unified Java and TypeScript codebase with Docker-backed local development workflows.

Big-Brain Live | *Python, PostgreSQL, REST APIs, GCP, FAISS*
- Engineered a cloud data-ingestion and indexing pipeline processing 50,000+ records with incremental updates, metadata persistence, and performance tuning, reducing retrieval latency by 40% across scalable document-search services.

FinNexus Live | *Next.js, TypeScript, Kubernetes, Plaid*
- Built a scalable fintech platform for 50,000+ users and 1M+ transactions, combining secure APIs, MongoDB data access, real-time updates, and Kubernetes-backed deployment workflows for reliable customer-facing operations.